

Emmie Kao

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EDUCATION

STANFORD UNIVERSITY

2025-2029

Intended B.A in International Relations; Data Science and Social Systems – Expected Graduation: 2029.

Relevant Coursework: Programming Abstractions, Statistical Methods for Social Scientists, Principles of Data Science.

THE BISHOP'S SCHOOL

2016-2025

Cum Laude Honoree — GPA: 4.61/4.0 — SAT: 1580, ACT: 36.

RESEARCH EXPERIENCE

SIMONS RESEARCH FELLOW INTERN, STONY BROOK UNIVERSITY DEPT OF COMPUTER SCIENCE

JULY-AUGUST 2024

- Applied Bayesian Additive Regression Trees (BART) and new Pattern Miner software to county-level data on the nationwide opioid epidemic to evaluate the effectiveness of laws.
- Demonstrated that the Pattern Miner reliably detected at-risk demographic subgroups and significant causal patterns in multidimensional data, outperforming BART in robustness to real-world data complexity.
- Delivered findings at the Simons Summer Research Symposium, providing evidence to support development of county-specific, data-informed public health policies.

Intern, University of San Diego Biophysics Lab

JUNE-AUGUST 2023

- Constructed a database for 10+ person lab using Python, HTML, CSS, and MySQL to organize lab materials.
- Coauthored a paper (<https://arxiv.org/abs/2411.02314>) on improving a form of image analysis using machine learning. Title: *Convolutional neural networks applied to differential dynamic microscopy reduces noise when quantifying heterogeneous dynamics.*

LEADERSHIP EXPERIENCE

OUTREACH INTERN, STANFORD WOMEN IN COMPUTER SCIENCE

2025-PRESENT

- Design and prepare an original winter-quarter workshop through biweekly board meetings.
- Plan biweekly Python programming classes attended by up to 100 elementary-school students.

COCHAIR, STANFORD MODEL UNITED NATIONS CONFERENCE

2025-PRESENT

- Co-lead a 60-person African Union committee over a three-day international conference for high schoolers.

MUSIC AND LOGISTICS INTERN, COUNTERPOINT A CAPPELLA

2025-PRESENT

- Rehearse 7 hrs/week, serve as musical director once every other week, plan group retreats and trips.
- Perform biweekly, both locally and internationally.

FOUNDER, DATA SCIENCE FOR CHANGE

2022-2025

- Educated students about data science and data-informed policymaking through a 501(c)(3) nonprofit.
- Led meetings and conducted outreach to establish national and international branches.

SKILLS & INTERESTS

Programming: Python (Pandas, Scikit-Learn, Tensorflow, Pytorch), C++, Java, Javascript, HTML, CSS, MySQL, Django.

Languages: Chinese (Intermediate Second-Year)

Awards: 16th in State for Impromptu Speaking, 5x Model United Nations Best Delegate.

Interests: Journalism (high school Editor-in-Chief of three publications), Music (singer & songwriter).
